COS AND COSH ON ONE PAGE

Circle and hyperbola, wave and chain

One exponential, averaged with its mirror image along the imaginary axis or along the real one.

THE RULE

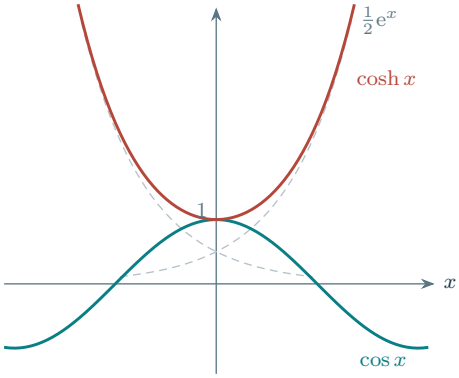
Average e^{z} and e^{-z}

$$\cos x = \frac{e^{ix}+e^{-ix}}{2}, \qquad \cosh x = \frac{e^{x}+e^{-x}}{2}, \qquad \cos(ix) = \cosh x$$

Euler's formula $e^{ix} = \cos x + i \sin x$ makes $\cos x$ the real part of a point going round the unit circle, and $\cosh x$ is the even part of e^{x}, the same average taken along the real axis. Their series differ only in sign,

$$\cos x = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \dots, \qquad \cosh x = 1 + \tfrac{x^2}{2!} + \tfrac{x^4}{4!} + \dots,$$

so $\cos$ solves $y'' = -y$ and oscillates, while $\cosh$ solves $y'' = y$ and grows; at a general complex point the two mix, $\cos(x+iy) = \cos x \cosh y - i \sin x \sinh y$.



Both start at 1 with zero slope; $\cosh$ is the average of the dashed curves.

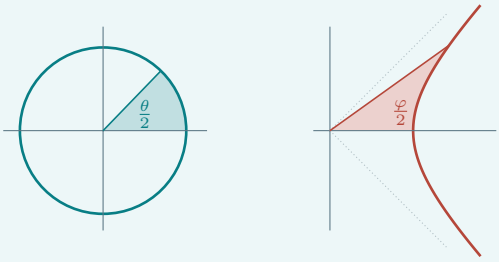
GEOMETRY

Twice the sector area, on two curves

On $x^2 + y^2 = 1$ the point $(\cos\theta, \sin\theta)$ cuts off a sector of area $\frac{\theta}{2}$, and on $x^2 - y^2 = 1$ the point $(\cosh\varphi, \sinh\varphi)$ cuts off a hyperbolic sector of area $\frac{\varphi}{2}$; the same definition of angle gives

$$\cos^2\theta + \sin^2\theta = 1, \qquad \cosh^2\varphi - \sinh^2\varphi = 1.$$

The circular angle returns after 2π, while the hyperbolic angle runs on without limit along a branch that never closes.



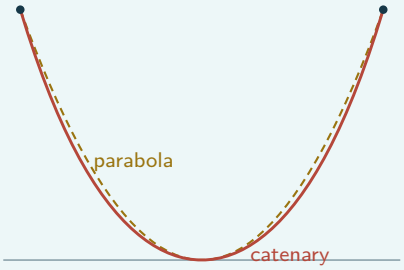
PHYSICAL SHAPES

A wave on a string, a chain hanging from it

A taut string obeys the wave equation, and its simplest solutions are travelling cosines $y = A\cos(kx - \omega t)$ with $\omega = ck$. The same string at rest under its own weight hangs as a catenary,

$$y = a\cosh\frac{x}{a}, \qquad y'' = \frac{1}{a}\sqrt{1+y'^2},$$

where a is the ratio of horizontal tension to weight per unit length. Inverted, the catenary is the arch that carries its own weight in pure compression, the shape behind Gaudí's hanging models and, weighted, the St Louis Gateway Arch; a parabola through the same ends sags differently because it carries load spread evenly across the span.



Same ends and lowest point: the chain is flatter at the bottom.

FOURIER ANALYSIS

cos builds signals, cosh controls growth

Because $\cos$ repeats, the functions $\cos nx$ are orthogonal on $[-\pi, \pi]$, with $\int \cos mx \cos nx \, dx = 0$ for $m \neq n$, so any reasonable periodic signal is a sum $\frac{a_0}{2} + \sum(a_n \cos nx + b_n \sin nx)$. $\cosh$ never repeats and cannot serve as such a basis; it enters instead when the exponent turns real, as the Laplace transform e^{-st} becomes the Fourier transform at $s = i\omega$. Separating Laplace's equation for a potential in a rectangle uses both at once, $u = \sin\frac{n\pi x}{L}\sinh\frac{n\pi y}{L}$, oscillating across and growing up.

BOOSTS

Rotations turn angles, boosts turn rapidities

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad \begin{pmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{pmatrix}$$

A rotation keeps $x^2 + y^2$ and a Lorentz boost keeps $c^2t^2 - x^2$, with $\gamma = \cosh\varphi$ and $v/c = \tanh\varphi$. Rapidities add where speeds do not: $v = \frac{3}{5}c$ has $\varphi = \ln 2$ and $\gamma = \frac{5}{4}$, two such boosts give $\varphi = \ln 4$ and $v = \frac{15}{17}c$, and the Doppler factor is simply $e^{\varphi} = 2$.

Conventions. $\sinh x = \frac{1}{2}(e^{x} - e^{-x})$ and $\sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$; angles are in radians, φ is the hyperbolic angle or rapidity, and $\tanh\varphi = \sinh\varphi / \cosh\varphi$.

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